

# Chap1\_Intro

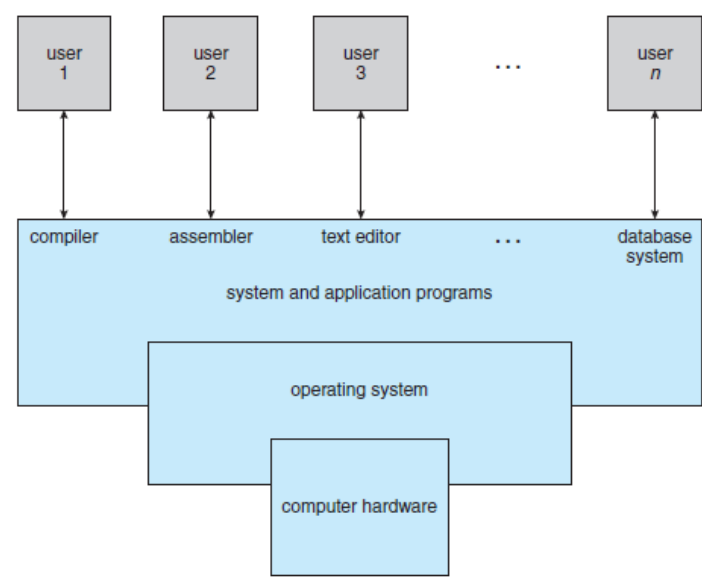
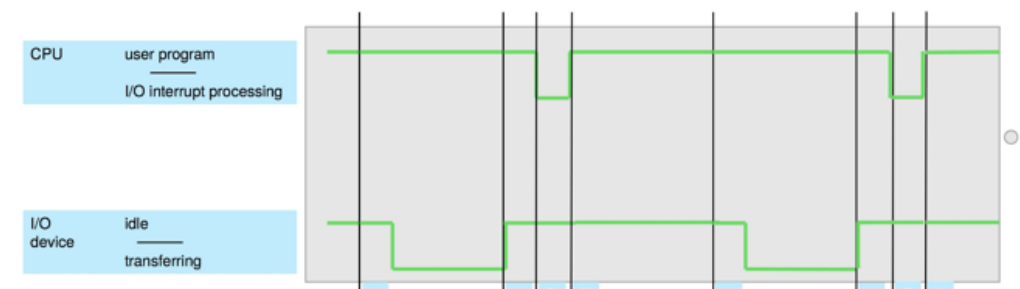


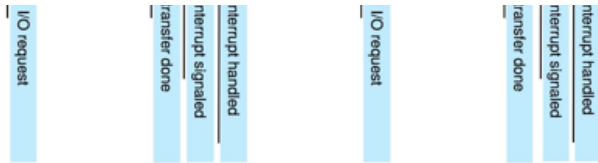
Figure 1.1 Abstract view of the components of a computer system.

## IO Structure

The matter of what constitutes an operating system became increasingly important as personal computers became more widespread and operating systems grew increasingly sophisticated. In 1998, the United States Department of Justice filed suit against Microsoft, in essence claiming that Microsoft included too much functionality in its operating systems and thus prevented application vendors from competing. (For example, a Web browser was an integral part of the operating systems.) As a result, Microsoft was found guilty of using its operating-system monopoly to limit competition

- Interrupt transfers control to the interrupt service routine generally, through the **interrupt vector**, which contains the addresses of all the service routines
- Interrupt architecture must save the address of the interrupted instruction
- A **trap** or **exception** is a software-generated interrupt caused either by an error or a user request
- An operating system is **interrupt driven**





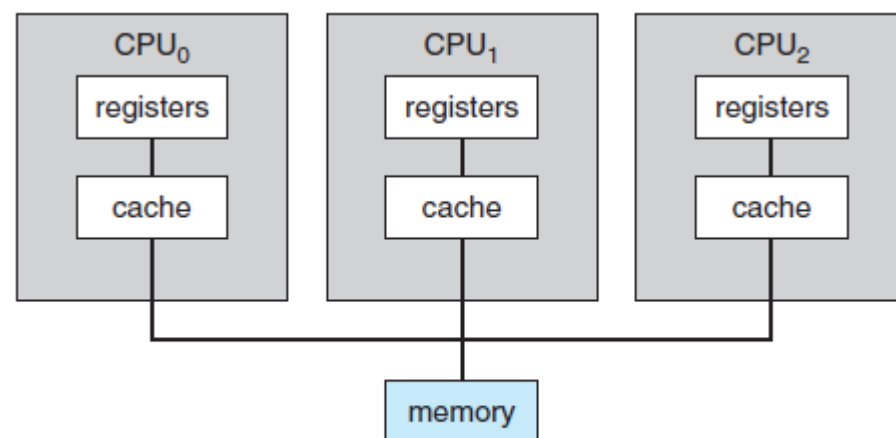
## Two methods for handling I/O

- After I/O starts, control returns to user program only upon I/O completion
  - Wait instruction idles the CPU until the next interrupt
  - Wait loop (contention for memory access)
  - At most one I/O request is outstanding at a time, no simultaneous I/O processing
- After I/O starts, control returns to user program without waiting for I/O completion
  - **System call** – request to the OS to allow user to wait for I/O completion
  - **Device-status table** contains entry for each I/O device indicating its type, address, and state
  - OS indexes into I/O device table to determine device status and to modify table entry to include interrupt

## SMP vs Asymmetric MultiProcessing

The multiple-processor systems in use today are of two types. Some systems use **asymmetric multiprocessing**, in which each processor is assigned a specific task. **Aboss** processor controls the system; the other processors either look to the boss for instruction or have predefined tasks. This scheme defines a boss-worker relationship. The boss processor schedules and allocates work to the worker processors.

The most common systems use **symmetric multiprocessing (SMP)**, in which each processor performs all tasks within the operating system. SMP means that all processors are peers; no boss-worker relationship exists between processors.



**Figure 1.6** Symmetric multiprocessing architecture.

of memory addressable in the system. Either way, multiprocessing can cause a system to change its memory access model from uniform memory access (UMA) to non-uniform memory access (NUMA). UMA is defined as the situation in which access to any RAM from any CPU takes the same amount of time. With NUMA, some parts of memory may take longer to access than other parts, creating a performance penalty. Operating systems can minimize the NUMA penalty through resource management, as discussed in Section 9.5.4.

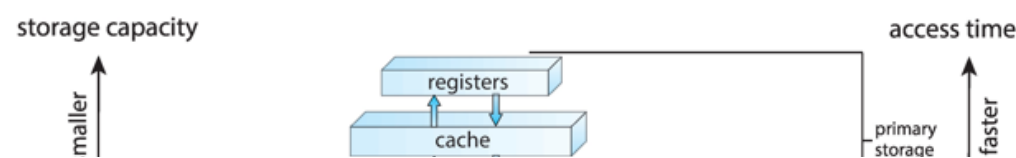
Clustered Systems

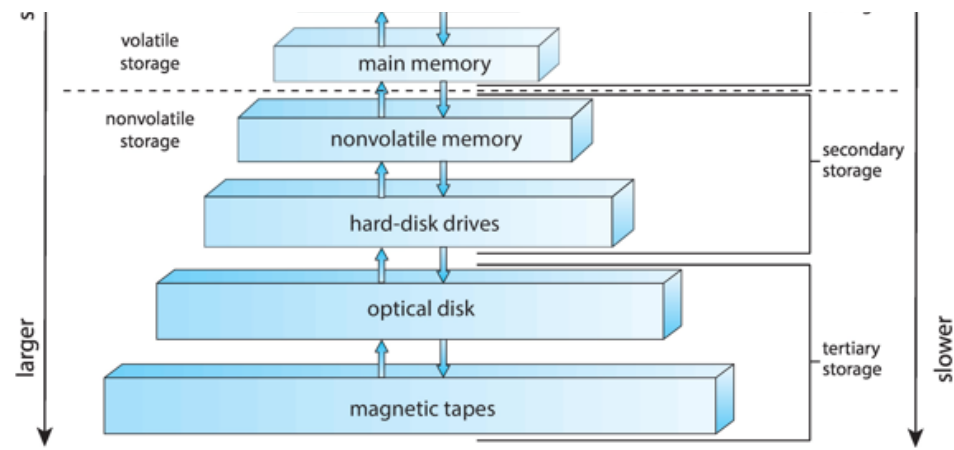
Clustering can be structured asymmetrically or symmetrically. In **asymmetric clustering**, one machine is in **hot-standby mode** while the other is running the applications. The hot-standby host machine does nothing but monitor the active server. If that server fails, the hot-standby host becomes the active server. In **symmetric clustering**, two or more hosts are running applications and are monitoring each other. This structure is obviously more efficient, as it uses all of the available hardware.

The application must have been written specifically to take advantage of the cluster; however. This involves a technique known as **parallelization**, which divides a program into separate components that run in parallel on individual computers in the cluster. Typically, these applications are designed so that once each computing node in the cluster has solved its portion of the problem, the results from all the nodes are combined into a final solution

Parallel clusters allow multiple hosts to access the same data on shared storage. Because most operating systems lack support for simultaneous data access by multiple hosts, parallel clusters usually require the use of special versions of software and special releases of applications. For example, Oracle Real Application Cluster is a version of Oracle’s database that has been designed to run on a parallel cluster. Each machine runs Oracle, and a layer of software tracks access to the shared disk. Each machine has full access to all data in the database. **distributed lock manager (DLM)**,

Storage Structure

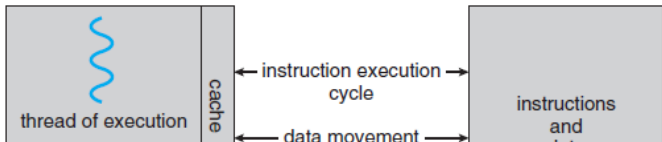




Depending on the controller, more than one device may be attached. For instance, seven or more devices can be attached to the **small computer-systems interface (SCSI)** controller. A device controller maintains some local buffer storage and a set of special-purpose registers. The device controller is responsible for moving the data between the peripheral devices that it controls and its local buffer storage. Typically, operating systems have a **device driver** for each device controller. This device driver understands the device controller and provides the rest of the operating system with a uniform interface to the device.

To start an I/O operation, the device driver loads the appropriate registers within the device controller. The device controller, in turn, examines the contents of these registers to determine what action to take (such as “read a character from the keyboard”). The controller starts the transfer of data from the device to its local buffer. Once the transfer of data is complete, the device controller informs the device driver via an interrupt that it has finished its operation. The device driver then returns control to the operating system, possibly returning the data or a pointer to the data if the operation was a read. For other operations, the device driver returns status information.

This form of interrupt-driven I/O is fine for moving small amounts of data but can produce high overhead when used for bulk data movement such as disk I/O. To solve this problem, **direct memory access (DMA)** is used. After setting up buffers, pointers, and counters for the I/O device, the device controller transfers an entire block of data directly to or from its own



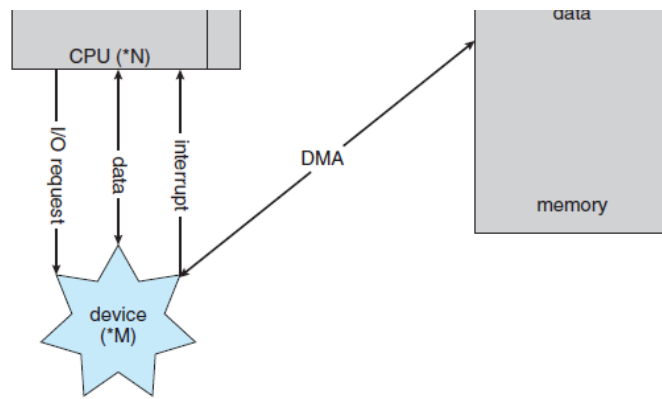
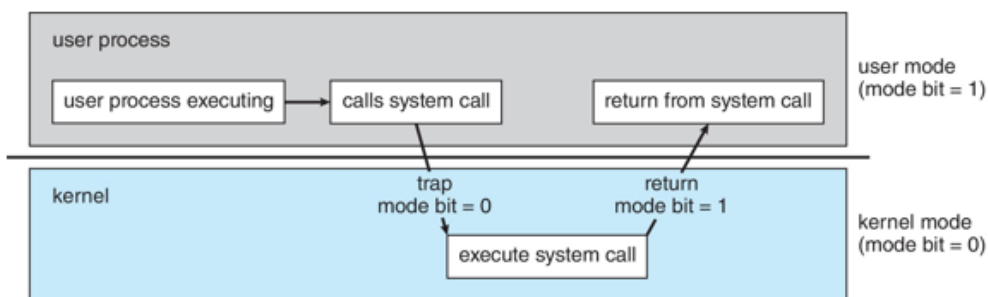


Figure 1.5 How a modern computer system works.

- Used for high-speed I/O devices able to transmit information at close to memory speeds
- Device controller transfers blocks of data from buffer storage directly to main memory without CPU intervention
- Only one interrupt is generated per block, rather than the one interrupt per byte

## Dual Mode operations

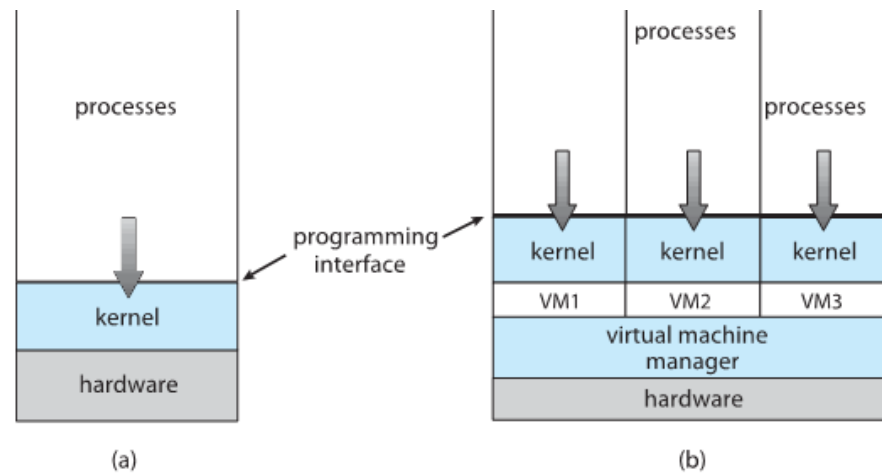


## OS Subsystems

- One purpose of OS is to hide peculiarities of hardware devices from the user
- I/O subsystem responsible for
  - Memory management of I/O including buffering (storing data temporarily while it is being transferred)
  - Caching (storing parts of data in faster storage for performance)
  - **Spooling (the overlapping of output of one job with input of other jobs)**
- General device-driver interface
- Drivers for specific hardware devices

## Virtualization





## Distributed Systems

- Collection of separate, possibly heterogeneous, systems networked together
  - **Network** is a communications path, **TCP/IP** most common
    - **Local Area Network (LAN)**
    - **Wide Area Network (WAN)**
    - **Metropolitan Area Network (MAN)**
    - **Personal Area Network (PAN)**
- **Network Operating System** provides features between systems across network
  - Communication scheme allows systems to exchange messages
  - Illusion of a single system

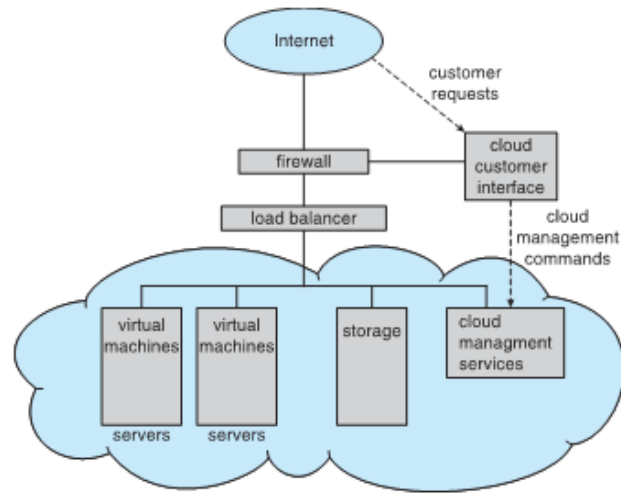
## Computing Environments

- Traditional
- Mobile
- Client Server
- Peer-to-Peer
- Cloud computing
  - Delivers computing, storage, even apps as a service across a network
  - Logical extension of virtualization because it uses virtualization as the base for its functionality.
    - Amazon **EC2** has thousands of servers, millions of virtual machines, petabytes of storage available across the Internet, pay based on usage
  - Many types
    - **Public cloud** – available via Internet to anyone willing to pay
    - **Private cloud** – run by a company for the company's own use
    - **Hybrid cloud** – includes both public and private cloud components
    - Software as a Service (**SaaS**) - word processor



- Platform as a Service (**PaaS**) – software stack ready for application use via the Internet (i.e., a database server)
- Infrastructure as a Service (**IaaS**) – storage available for backup use

◦



- Real-time Embedded
  - Real-time embedded systems most prevalent form of computers
  - Vary considerable, special purpose, limited purpose OS, **real-time OS**

## Protection vs Security

- Protection refers to keeping data intact from other subsystems malfunctioning.
- Security refers to the authentication of users who are allowed to RWE a file.

## Kernal Data structures

- Linked Lists
  - Singly
  - Doubly
  - Circularly
- Trees
- HashMaps
- Bitmaps - store the bit indicating the resource availability. Can be used when large data needs to be represented.

☐ DO this

## LINUX KERNEL DATA STRUCTURES

The data structures used in the Linux kernel are available in the kernel source code. The *include* file `<linux/list.h>` provides details of the linked-list data structure used throughout the kernel. A queue in Linux is known as a `kfifo`, and its implementation can be found in the `kfifo.c` file in the kernel directory of the source code. Linux also provides a balanced binary search tree implementation using *red-black trees*. Details can be found in the

```
include file <linux/rbtree.h>.
```